Raft and Fault-Tolerant Key/Value Storage: Consensus and State Machine Replication

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***Abstract*—[TODO: 150–200 words: problem, algorithms (leader election and consensus/replication), containerized environment, metrics, and the main result (linearizable under faults; latency cost under loss/delay).]**

I. Introduction

Replicating a service across several machines is the standard way to make systems resilient to machine crashes, but replication alone is not enough: the replicas must agree on a single order of operations, and they must keep agreeing while links drop messages, nodes leave and reenter the grid, leaders disappear, among other failures. Many consensus algorithms that provide a solution to this problem exist today, and maintaining a log of agreed-upon state-machine operations is a common implementation approach. The Raft algorithm [1] is a widely used consensus protocol that is designed to be understandable and easy to teach, and it is the basis of many production systems, including etcd, Consul, and CockroachDB **[Everton:** cite]. The practical question, however, is not only whether a protocol can be implemented correctly, but what that correctness costs in throughput, tail latency, and recovery time when faults actually happen.

With this in mind, this project had two main elements: 1. Implementing Raft, as well as a simple key-value store system on top of it, with the goal of understanding in detail how this consensus algorithm works; and 2. Building an experimental harness on top of these implementations to measure the cost of fault tolerance in a realistic environment. By implementing the resulting system, key distributed systems concepts were explored, such as leader election, consensus, replication and consistency, fault detection and tolerance, peer-to-peer communication, among others.

Basis and contributions

The starting point for this project was the framework supplied by MIT's 6.824 (2020) course labs 2 and 3 [2]. It provided stub APIs, an in-memory simulated network (`labrpc`), a gob (Go's serialization library) wrapper (`labgob`), the Porcupine linearizability checker [3], and, crucially, a comprehensive and thorough set of tests. This skeleton was kept intact and was used to guide and test our implementation; the protocol and service implementations, the runtime transport, persistence, container packaging, observability, and the whole experimental apparatus were written by us. Concretely, this project's contributions are:

- **In-House Raft Implementation:** following MIT 6.824's labs 2A–2C + 3B, our custom raft implementation closely follows Raft's extended paper guidelines, including leader election with randomized timeouts, the log up-to-date voting rule, log replication with fast backup, safety rules for commitment across terms, persistent state restored after a crash, and log compaction with snapshots.
- **Fault-tolerant Key/Value Service Implementation:** following MIT 6.824's labs 3A–3B, a replicated key-value store service was built on top of the Raft implementation built before. It includes client-side leader discovery with retry, response caching, and state snapshotting.
- **Deployment Setup**: the systems' transport layer, ports, and persistence behavior were made configurable via environment variables, so that the same code can run both in the labs' test suites (over the simulated `labrpc` network) and in a containerized deployment (over real `net/rpc`/TCP). A Dockerfile for container image configuration, and a `compose.yaml` file for setting up multiple nodes were provided.
- **Interface and Experimental Harness**: an HTTP/JSON API was exposed so that clients can interface with the KV application. A `/status` endpoint was also created, exposing term, leader, commit index, last log index, applied index and /snapshot sizes, plus a benchmark client and a scenario-driven experiment driver (normal operation, network delay/loss, leader crash, partition, cluster-size sweep) that record raw CSVs and check every history for linearizability with Porcupine.

[TODO: Check that this contribution list matches the emphasis of the abstract and of the five-minute video, and trim it if the six-page limit becomes tight.**]**

[Everton: This next paragraph needs review..**]** The experiments show that the service remains linearizable in every scenario we exercised, including 2% packet loss, leader crashes and partitions, while making the cost of fault tolerance explicit: throughput drops by about two orders of magnitude under 50 ms delay with 2% loss (**[Everton:** this seems awfully high, verigy**]**), failover takes roughly the election timeout, and the client-visible recovery time is dominated by the client's own RPC deadline rather than by the protocol.

[TODO: Confirm the numbers in the previous paragraph once the final experimental batch is frozen, and keep them consistent with Table II and Table III.**]**

The remainder of this report is organized as follows. Section II states the system model and the failure assumptions. Section III describes the architecture and the container

deployment. Section IV details the two algorithms and lists the reused code and our modifications. Section V argues the correctness properties and describes the automated tests. Section VI defines the experimental methodology and its reproducibility, Section VII presents the results, Section VIII discusses limitations and threats to validity, and Section IX concludes.

II. System Model

[Everton: Essa seção faz muitas referencias ao funcionamento do raft, sem explica-las antes. Verificar com o professor se uma explicação prévia do raft é necessária em uma possível seção "Background"**]**

We consider a fixed set of N servers that talk to each other, plus any number of clients. These server nodes are both KV-store servers and Raft peers. Clients come and go and only send requests. We make no assumptions about how fast nodes run or about shared clocks. The assumptions listed here are the ones we rely on when arguing correctness and progress in Section V.

A. Timing model

Raft is partially synchronous. It does not know in advance any hard limit on message delay or on how long a node takes to execute a step, but timeouts and heartbeats managed by local timers are used to detect failures and guarantee progress.

Safety does not depend on timing at all. The rules for electing a leader, matching logs and committing entries use only the contents of messages and the fact that majorities intersect, so no timeout can make two replicas disagree. Progress does depend on timing, and on the constants in Table I: heartbeats every 150 ms, election timeouts of 500–700 ms, peer RPCs bounded at 800 ms and client requests at 2 s.

B. Channels

Each pair of nodes has its own connection, and messages are addressed to one specific node and there is no broadcast medium. Raft communication happens via remote procedure calls, and client-server interaction with the KV-store service layer is done with HTTP requests. We assume that a channel can lose, delay, duplicate and reorder messages, and that a message may take arbitrarily long or never arrive. Retransmission and retries are our job, not the channel's. In the deployed system the connection is TCP carrying `net/rpc` with gob encoding, and we inject loss and delay at the application level; in the test suite the same protocol runs over `labrpc`, which already simulates loss, reordering and delay. Since requests and replies can be duplicated, every operation is idempotent: clients tag each request with a `(ClientId, Seq)` pair and servers answer repeated requests from a cache instead of applying them again.

C. Failures

It is assumed that a node can crash and later restart, losing its volatile state but keeping what it wrote to disk. The network can also split into partitions, for a while or forever. We do not consider Byzantine faults; messages are not forged or corrupted, apart from what the serialization library's decoding can detect. Raft guarantees correctness in (and thus limits progress to) situations in which, with N replicas, no more than $\lfloor (N-1)/2 \rfloor$ nodes fail concurrently.

D. Clocks, persistence and restart

Raft does not assume synchronized clocks, and no algorithm step reads wall-clock time; a local monotonic clock is used only to measure timeouts. Before a node replies to a request or acts on one, it writes to disk what it needs after a crash: the current term, the vote, the log and the last included snapshot index. Each write goes to a temporary file that is then renamed, so a crash in the middle of a write cannot leave truncated state. On restart the node reloads those files and resets only its volatile state (role, leader hint and timers), coming back as a follower.

E. Correctness properties

The system is correct when two kinds of properties hold:

- *Safety:* there is at most one leader per term; two logs that share an entry at the same index also share the whole prefix before it (Log Matching); an entry committed at one replica eventually appears, with the same index and command, at every replica; and the KV service is linearizable, meaning that every read and write looks atomic, taking effect at some instant between its call and its reply.
- *Liveness:* once the network is stable and a majority of nodes is up, a leader is elected after finitely many elections and every command submitted to it is eventually committed and applied.

Section V argues safety and describes the tests that exercise it.

III. Architecture

[TODO: Figure/diagram: HTTP client → nodes (API :9000) and Raft mesh (:8000) between containers; persistent volumes.**]** **[TODO:** Describe `compose.yaml`: 3+ nodes, bridge network, ports 9000–9002 published, 8000 internal, one volume per node.**]** **[TODO:** Communication: binary RPC (`net/rpc`/gob) between peers; HTTP/JSON API for clients; `Clerk` on the client with retry and leader switching. Explain why the client (not the server) finds the leader.**]** **[TODO:** Observability: `/status` endpoint (term, leader, indexes, state and snapshot sizes) used by the experiments.**]** **[TODO:** Versions: Go, Docker/Compose, base image — fill in Table I.**]**

IV. Implemented Algorithms

[TODO: Make explicit that these are the two complementary algorithms: leader election (theme 1) and consensus/replication (theme 2).**]**

A. Leader election and terms

[TODO: Randomized timeouts, voting (log up-to-date rule, one vote per term), conversion to follower on a higher term; role of the observed `leaderId`.**]**

Table I
EXECUTION ENVIRONMENT

Item	Value
Hardware	[TODO: CPU, cores, RAM]
Operating system	[TODO: distribution, kernel]
Docker / Compose	[TODO: versions]
Go	[TODO: version]
Nodes	3, 5, 7 containers (same host)
Load	8 clients, 50 ops/client
Seeds	5–10 per configuration
Warm-up	1 discarded run (sweep)

B. Log replication and consensus

[TODO: `AppendEntries` with `PrevLogIndex/Term`, retry with `XTerm/XIndex/XLen`, `commitIndex` advanced only for entries from the current term (Figure 8 of the paper), snapshots and `InstallSnapshot`.]

C. Linearizable KV state machine

[TODO: Every operation (reads included) goes through the log; response cache keyed by `(ClientId, Seq)` for exactly-once application; replies matched by log index; snapshot of the map plus the cache.]

D. Reused code and modifications

[TODO: Mandatory citation: MIT 6.824 labs 2A–3B skeleton (`labrpc`, `labgob`, `porcupine`, tests) [2]; describe what was changed: the Raft/KV implementation, the `RPCClient` interface for real TCP, the on-disk `Persister`, the `/status` endpoint, the benchmark client and the experiment scripts.]

V. CORRECTNESS PROPERTIES AND TESTS

A. Safety arguments

[TODO: Sketch the proof: leader uniqueness per term; majority intersection; the election restriction guarantees the elected leader holds every committed entry; in-order application by index.]

B. Automated tests

[TODO: Describe the inherited suite (2A/2B/2C, 3A/3B) and our own tests: real TCP cluster, follower rejecting writes (HTTP 421), `/status` consistency. Report that everything passes (total time).] [TODO: Linearizability: Porcupine reconstructs a valid order for the history; 46/46 runs `linearizable=ok` in the experimental scenarios.]

VI. EXPERIMENTAL METHODOLOGY

[TODO: Describe the workload (50/50 read/write mix, 5 keys, `kvbench`), the per-run key namespace, and the fact that throughput is measured before the linearizability check.] [TODO: Metrics: throughput (ops/s), latency p50/p95/p99, time-to-elect, service recovery time (TTR-service), replica recovery time (TTR-replica), attempts per operation and `ErrWrongLeader` rate. Justify each one.] [TODO: Parameters varied at ≥ 3 levels: $N \in \{3, 5, 7\}$; delay/loss via `tc netem` (0, 50 ms, 100 ms; 0%, 2%, 5%); number of clients.] [TODO: Fault scenarios: leader crash, leader partition, full cluster restart.] [TODO: Reproducibility: `scripts/experiment.py` with recorded seeds, raw CSVs in `scripts/experiments/`, analysis in `scripts/analysis.ipynb`. Exact commands.]

Table II
THROUGHPUT AND LATENCY (MEDIAN OVER SEEDS)

Scenario	ops/s	p50	p95	p99
Normal	1633	4.5	7.4	8.8
`netem` 50 ms/2%	41	155	307	418
Leader crash (2 nodes)	2038	2.7	8.0	21.9
Partition (after heal)	1227	2.6	6.4	184

Table III
RECOVERY TIMES (MEAN ± STD)

Scenario	election	TTR-service	TTR-replica
Leader crash	507 ± 114 ms	1868 ± 32 ms	468 ± 124 ms
Partition	349 ± 185 ms	1632 ± 404 ms	not applicable

VII. RESULTS

[TODO: Replace the values below with the final version and generate the figures with `scripts/analysis.ipynb`.]

A. Normal operation

[TODO: Describe throughput and percentiles; explain that the 2% retry in the baseline is one extra attempt per client to discover the leader (8/400), not steady-state retrying.]

B. Delay and loss

[TODO: Discussion: throughput drop and p99 increase; why the ceiling is dominated by the election timeout and retransmissions; linearizability is preserved.]

C. Leader crash

[TODO: Time-to-elect $\approx$ election timeout; TTR-service dominated by the client HTTP timeout ($\sim$2 s); TTR-replica measures recovery via log/snapshot.]

D. Partition and the disruptive-server problem

[TODO: Inflated terms, churn, attempts per operation; explain that without PreVote the isolated node exiles itself; treat it as a limitation and future work.]

E. Scalability

[TODO: Results with $N = 3, 5, 7$: throughput and latency; discuss why the leader serializes and where replication cost would show up (message counts).]

VIII. LIMITATIONS AND THREATS TO VALIDITY

[TODO: Containers on one host are not independent failure domains; variability from short runs; no batching; no PreVote; no message/byte counters; Docker Desktop environment.]

IX. Conclusion and Future Work

[TODO: Recap: two algorithms, correctness under faults, performance cost. Future work: PreVote/check-quorum, batching, lease-based reads, incremental snapshots, message accounting, execution across multiple machines.]

References

[1] D. Ongaro and J. Ousterhout, "In search of an understandable consensus algorithm," in *2014 USENIX Annual Technical Conference (USENIX ATC 14)*. Philadelphia, PA: USENIX Association, 2014, pp. 305–319. [Online]. Available: https://www.usenix.org/conference/atc14/technical-sessions/presentation/ongaro

[2] R. Morris, F. Kaashoek *et al.*, "6.824 distributed systems: Labs 2 and 3," MIT Computer Science and Artificial Intelligence Laboratory, 2020, skeleton for `labrpc`, `labgob`, `porcupine` and the test suite reused in this work. [Online]. Available: https://pdos.csail.mit.edu/6.824/2020

[3] A. Athalye, "Porcupine: A fast linearizability checker for go," GitHub repository, 2017. [Online]. Available: https://github.com/anishathalye/porcupine